

DS type Mines

I. Thème : traduire en anglais le texte ci-dessous (8 points)

Après la Black Barbie, lancée avec sa coupe afro dans les années 1980, le fabricant américain de jouets Mattel brise un tabou en annonçant, jeudi 28 janvier, de multiples déclinaisons de sa Barbie. Elle ne sera plus seulement incroyablement longiligne comme aujourd'hui, mais également commercialisée dans trois nouvelles versions : « ronde », « petite » ou « grande ».

« *Barbie est le reflet du monde que les jeunes filles voient autour d'elles* », a expliqué Richard Dickson, le PDG de Mattel. Il était temps. En cinquante-six ans, Barbie - cette poupée sexy aux mensurations irréalistes, imaginée par Ruth Handler, l'une des fondatrices de Mattel, était devenue un concentré de stéréotypes féminins.

« *Nous avons la responsabilité, envers les filles et les parents, de refléter une vue plus large de la beauté* », a assuré Evelyn Mazzocco, vice-présidente et manager générale de la marque. Ce virage arrivera-t-il à temps pour [relancer](#) les ventes, en chute libre depuis trois ans, du fabricant de jouets ? La fonte des bénéfices du groupe a déjà coûté sa place, fin 2014, à l'ancien PDG Bryan Stockton.

Le Monde, 28 janvier 2016.

Mapping the uncanny valley: why androids are scary

ARTIFICIALLY created beings, whether they be drawn or sculpted, are warmly accepted by viewers when they are distinctively inhuman. As their appearances are made more real, however, acceptance turns to discomfort until the point where the similarity is almost perfect, when comfort returns. This effect, called "the uncanny valley" because of the dip in acceptance between clearly inhuman and clearly human forms, is well known, particularly to animators, but why it happens is a mystery. Some suggest it is all about outward appearance, but a study just published in *Cognition* by Kurt Gray at the University of North Carolina and Daniel Wegner at Harvard argues that there can be something else involved as well: the apparent presence of a mind where it ought not to be.

The scientists theorised that adding human-like eyes and facial expressions to robots conveys emotion where viewers do not expect emotion to be present. The resulting clash of expectations, they thought, might be where the unease was coming from.

To test this idea, the researchers presented 45 participants recruited from subway stations and campus dining halls in Massachusetts with a questionnaire about the "Delta-Cray supercomputer". A third were told this machine was "like a normal computer but much more powerful". Another third heard it was capable of experience, by being told it could feel "hunger, fear and other emotions". The remainder were told it was capable of "self-control and the capacity to plan ahead", thus suggesting it had agency. Participants were asked to rate how unnerved they were by the supercomputer on a scale where one was "not at all" and five was "extremely".

Dr Gray and Dr Wegner found that those presented with the idea of a supercomputer that was much more powerful than other computers or was capable of planning ahead were not much unnerved. They gave it a score of 1.3 and 1.4 respectively. By contrast, those presented with the idea of a computer capable of experiencing emotions gave the machine an average of 3.4. These findings are consistent with the researchers' hypothesis.

The Economist July 21st 2012 | from the print edition

Question 1: Explain to what extent Dr Gray and Dr Wegner's findings are consistent with their hypothesis. Answer the question in your own words. 70 to 120 words.

Question 2:

Does aesthetics matter in engineering? Do you think that engineers should receive training in applied arts? 110 to 200 words